

Object Detection for Rescue Operations

Project Report Submitted in fulfilment of the Requirements for the Award of the Internship of Winter Training Program on Space Science Technology

Advanced drone Technology

By

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**Under the Supervision of
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Abstract

In rescue missions, it is difficult to locate missing people due to extreme landscapes that are difficult for rescue teams to reach in person.

This vision-based real-time target acquisition system is specifically manufactured for search-and-rescue operations, using the help of AI with a pre-trained model of the YOLOv8 neural network.

It helps in filtering out the unnecessary objects recorded and puts a target lock on the specified objects as they reach the centre of the frame.

The drone HUD displays a target-locked box, the distance of the object from the drone, a live fps counter, and the confidence score.

Working Logic

The video was captured using OpenCV, which converts its singular frames into numeric pixel values and stores them in the form of a NumPy array.

Inside the function, the deep learning-based object detector, yolov8n.pt, was called to perform the necessary actions.

Bounding boxes are created, which act as a boundary to separate the specified objects from the remaining scenery.

The boxes locate the position of the object in (x,y) co-ordinates and classify each object into separate categories, which are already predefined into the model and show the confidence score of the detected object, on how certain the model is regarding its detection.

The confidence threshold is maintained at 0.6 so as to avoid catching any weak detections.

The centre of the box is calculated so that when an object reaches the middle 20% of the frame, the box turns red, stating that the target is locked. Otherwise, the box remains green, which is the scanning mode.

A live fps counter helps to detect slow detections, while the model keeps executing, so that performance remains stable in real time.

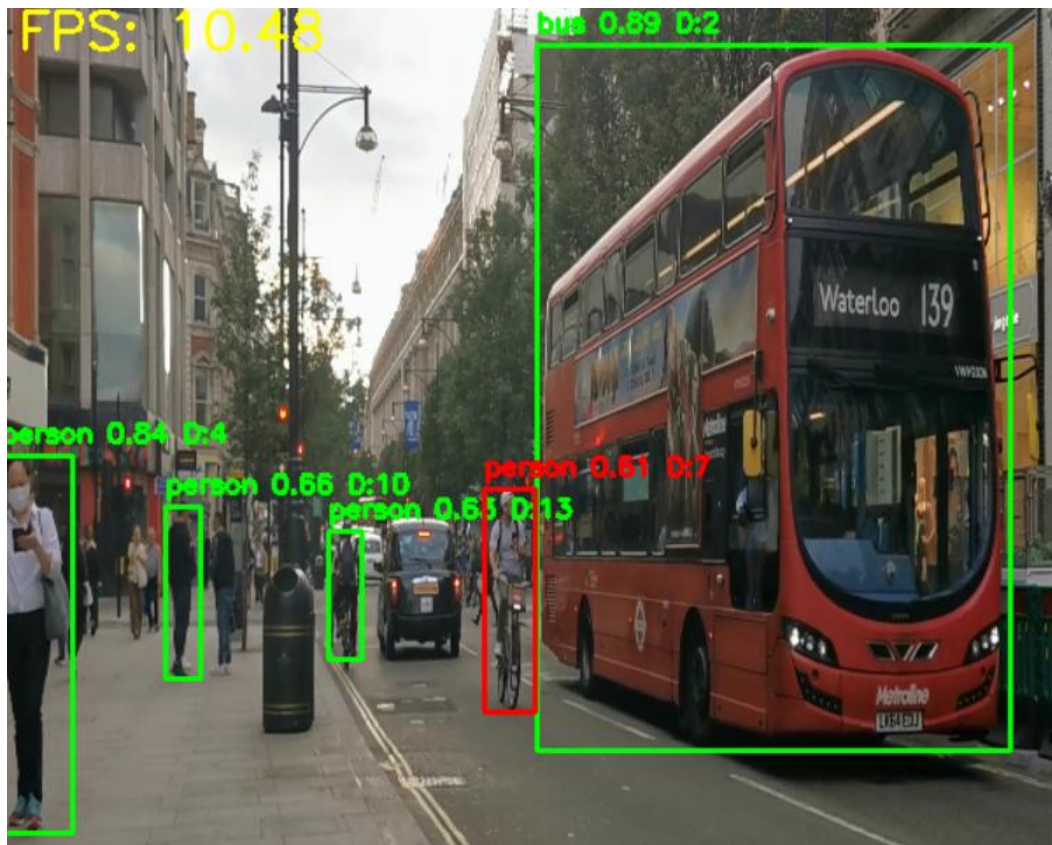
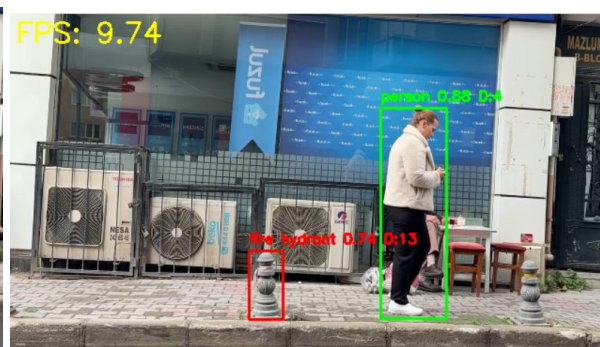
The system achieved 15–16 FPS on CPU execution at 640×640 resolution.

Code and Output Screenshots

The complete file is attached below:



telemetry system.zip



Computer Vision Foundations with OpenCV and NumPy

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Assignment-1

Landing Pad Finder

The required folder is added below:



Assignment_1.zip

Working screenshots:



Original



Output

Assignment-2

Horizon Lander

The required folder is added below:



Assignment_2.zip

Working screenshots:



Original



Output

Assignment-3

Obstacle Alert

The required folder is added below:



Assignment_3.zip

Working screenshots:



Original



Masked



Annotated

Assignment-4

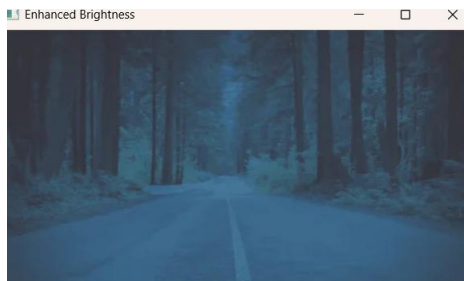
Night Vision

The required folder is added below:

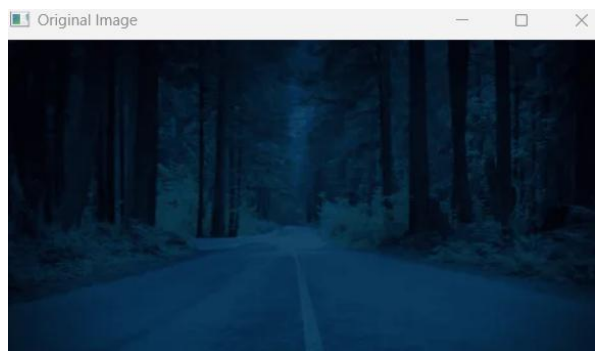


Assignment_4.zip

Working screenshots:



Original



Brightened

Assignment-5

Motion Blur

The required folder is added below:



Assignment_5.zip

Working screenshots:



Original



Blurred

